

COMICCRAFTAI

MODULE DESIGN

1. INTRODUCTION

ComicCraftAI is divided into multiple modules to make the application organized, maintainable, and easier to develop.

2. MAIN MODULE

File:
app/main.py

Purpose:

Creates the FastAPI application, configures static file serving, and includes the application routes.

3. ROUTES MODULE

File:
app/routes.py

Purpose:

Handles HTTP requests such as the home page, comic generation, image testing, and export-related routes.

4. OUTLINE GENERATION MODULE

File:
app/gemini_flash.py

Purpose:

Communicates with Gemini to generate a structured five-panel comic outline.

5. STORY GENERATION MODULE

File:
app/gemini_pro.py

Purpose:

Generates captions, narration, and dialogue for the comic panels.

6. IMAGE GENERATION MODULE

File:

app/image_generator.py

Purpose:

Sends image prompts to the Hugging Face inference service and saves the generated panel images.

7. LAYOUT BUILDER MODULE

File:

app/layout_builder.py

Purpose:

Combines the outline, story information, and generated images into a unified comic layout.

8. PDF EXPORT MODULE

File:

app/exporters.py

Purpose:

Creates the final PDF containing comic images and textual information.

9. FRONTEND MODULE

Files:

templates/index.html

templates/comic_preview.html

templates/export_success.html

Purpose:

Provides the user interface and displays the generated comic.

10. STATIC FILES

The static directory contains CSS files, generated panel images, and generated PDF exports.

11. CONCLUSION

The modular design separates the major responsibilities of ComicCraftAI and supports easier maintenance and future expansion.